

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

Nov 2013

EE303 Microcontrollers & Applications.

Date : 22-11-2013 (Friday)

Time: 10.00 am to 01.00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q-1 (a) List out peripherals available in 8051 microcontroller. 2
- (b) Explain difference between Micro Processor & Microcontroller. 5
- Q-2 (a) Explain Function of following pins of 8051 Microcontroller. 4
1. XTAL1 & XTAL2
 2. RST
 3. TXD & RXD
 4. INT0 & INT1
- (b) Draw Port Pin Structure for any one Port & explain the same. 6
- (c) Write a program to send 80H, 20H & 10H in sequence on Port 1. 4
- OR
- Q-2 (a) Draw internal Memory Architecture for 8051 microcontroller with information of address range for each segment of memory. 6
- (b) Write a program to continuously toggle the status of Port Pin P1.0. Use software delay loop between successive toggle of Port pin. 8
- Q-3 (a) Explain any four assembly instruction for Branching operation. 4
- (b) Explain Direct & Indirect addressing methods of data transfer with short example program. 5
- (c) What do you mean by interrupt? How interrupt acknowledgement is better than Pooling? List out various interrupts of 8051. 5
- OR
- Q-3 (a) Explain any four assembly instruction for Logical operation. 4
- (b) Give bit definition of PSW register & explain function of each bit of it. 4
- (c) Explain Assembly directives used in assembly language programming of microcontroller. 6

SECTION – II

- Q-4 (a) Explain difference between Timer mode of operation & Counter mode of operation. 2
- (b) List out any five application in which you have seen observed either Key board or LCD interfaced with microprocessor/microcontroller directly or indirectly. 5
- Q-5 (a) Draw & explain working of Timer0 in Auto Reload mode(Mode 2). 6
- (b) Write & explain a Program which demonstrates the use of Timer0 in Auto Reload mode. The Port pin P1.0 Should be toggled each time when timer overflows. 8

OR

- Q-5 (a) Explain Working of Timer0 in 16 bit Mode(Mode 1). 6
- (b) Write & Explain function of each bit in TMOD register. 4
- (c) Write & Explain function of each bit in TCON register. 4
- Q-6 (a) Draw interfacing scheme for Seven Segment LED display & 8051. 6
- (b) Explain interfacing of ADC0808 with 8051 microcontroller. 8

OR

- Q-6 (a) Find out number to be supplied to Common Anode & common cathode LED display for displaying 0 to 9. 8
- (b) Explain how would you count the bottle passing on conveyer using 8051 microcontroller. 6